

A Survey on Agentic AI Approaches for Unifying Knowledge Graphs and Large Language Models

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Abstract. The rapid progress of Large Language Models (LLMs) and Knowledge Graphs (KGs) has created new opportunities for building intelligent systems that are simultaneously expressive, grounded, and capable of complex reasoning. However, each paradigm has distinct limitations: LLMs struggle with factual consistency, long term memory, and controllability, while KGs lack the generative fluency and adaptive reasoning required for open ended tasks. This survey provides the first comprehensive study of how these two paradigms are increasingly unified through the emerging paradigm of agentic AI systems composed of autonomous agents. The survey introduces agentic AI as a conceptual and architectural bridge that unifies symbolic and neural intelligence, charting a path toward more adaptive LLM–KG agentic ecosystems. To the best of our knowledge, this is the first survey to explore the synergy between knowledge graphs and LLMs from an Agentic AI perspective.

Keywords: LLM · Knowledge Graph · Agentic AI.

1 Introduction

Knowledge Graphs (KGs) [29] organize entities and their relationships in a structured and semantically rich manner. They are verifiable and interpretable explicit

knowledge that helps in symbolic reasoning and proper querying and is therefore necessary in construction of reliable and explainable intelligent systems. The Large Language Models (LLMs) [33] turn out to be able to comprehend and generate natural language on a wide range of tasks and shown an outstanding performance across various diverse natural language processing tasks. When combined with KGs, LLMs gain access to structured knowledge that improves coherence, factual correctness, and interpretability. Conversely, KGs benefit from LLM-driven knowledge extraction, linking, and natural language explanations [28].

Despite their individual strengths, both LLMs and KGs face limitations when applied to increasingly complex tasks like multi-hop reasoning, realtime decision making, external tool use, etc. LLMs often struggle with factual grounding, hallucinations, and long-term memory, while KGs, though accurate and interpretable, lack the flexibility and generative capabilities of language models. Moreover, KGs require manual curation and schema implementation when it is of particular domains in specific. To address these challenges, recent research has embraced hybrid approaches, combining LLMs and KGs to create systems that are both generative and grounded, flexible yet factual. This synergy has shown promising results in tasks like question answering [46], retrieving information [41], evaluating knowledge [27], and conversational AI [48]. While several surveys have explored the integration of LLMs, KGs and its limitations [28, 52, 30, 80], gaps remain—particularly in the context of emerging paradigms such as agentic AI.

Recently, Agentic AI represents a significant evolution in the capabilities of intelligent systems. It refers to systems built around autonomous, goal-driven agents that can perceive, reason, plan, and act in complex environments—often leveraging tools like LLMs, KGs, APIs, databases, and even other agents to achieve their objectives. This survey aims to provide a fresh perspective on the evolving landscape of LLM-KG integration in the age of agentic AI as an extension of previous survey [28]. We not only revisit and categorize prior work on LLM-KG interplay but also examine how these technologies are being embedded into agentic systems that exhibit autonomy, memory, tool-use, reflection, and coordination capabilities. Furthermore, we propose a refined categorization of the LLM-KG relationship, extending it to account for the emerging paradigm of agentic AI. These interactions can be understood across three functional axes—LLMs for KGs, KG-enhanced LLMs, and LLM-KG Cooperation—and three agentic configurations, reflecting the architectural emphasis of the agent: LLM-only, KG-only, or hybrid systems. This categorization serves as a conceptual bridge between static LLM-KG integration and dynamic, task-performing intelligent agents.

The paper is organized as follows: Section 3 explores the role of LLMs within the context of KGs and Section 4 investigates how KGs can enhance LLMs. In Section 5, we examine the collaborative utilization of LLMs and KGs. Section 6 explore the identification of open challenges. Finally, we conclude in Section 7.

2 Literature Review

The synergy of LLMs [54] and Knowledge Graphs has gained a huge attention on research across different application domains. Recently, several approaches merged LLMs and KGs by relying mainly on the LLM’s query interpretation but failed to understand deep semantics and cognitive knowledge insight the KGs [64]. To address this issue, several AI agents are introduced as an autonomous LLM-driven framework designed to execute advanced reasoning tasks across the Knowledge Graphs. Overall, KGs provide trust and semantic consistency, LLMs offer cognitive enrichment, and the agent layer offers actionable decision making and coordination to form interpretable, robust and autonomous intelligent systems. In this section, some KG+LLM based agents such as ESCARGOT [47], KARMA [43], ODA [64] and KG-Agent [32] are discussed.

ESCARGOT [47], an AI agent that combines Large Language Models with a dynamic Graph of Thoughts and biomedical knowledge graphs to improve reasoning and reduce hallucinations. It generates executable strategies, uses Python-based structures, and retrieves verified knowledge through databases and queries. Compared to standard RAG systems, it performs better on complex biomedical questions while offering transparent and verifiable reasoning.

KARMA [43], is a multi-agent LLM framework designed to automate knowledge graph enrichment from scientific literature. It uses specialized agents for tasks like entity extraction, relation discovery, and schema alignment, enabling structured knowledge extraction from unstructured text. Its collaborative verification approach improves accuracy, reduces inconsistencies, and scales effectively across biomedical domains, achieving strong results on PubMed data with high correctness and fewer conflicts.

ODA [64], is an observation-driven agent that integrates LLM reasoning with knowledge graphs through an iterative observe–act–reflect process. It continuously extracts and reasons over relevant subgraphs, improving multi-hop reasoning and relation extraction. This approach demonstrates that actively leveraging KG structure significantly enhances LLM performance on complex tasks.

KG-Agent [32], an autonomous agent framework built to improve complex reasoning over knowledge graphs (KGs) by large language models (LLMs). Rather than simply retrieving facts, KG-Agent enables a relatively small LLM (e.g., a 7 B-parameter model) to actively decide which tools to apply, how to navigate the graph, and when to stop. The details are summarized in Table 1.

3 LLM for KG

LLMs support KG construction across multiple stages, particularly in ontology generation, entity/relation extraction, and text generation. For ontology creation, In entity extraction, prompt based and instruction tuned models such as PromptNER [2] and UniversalNER [84] improve domain specific NER with minimal supervision. For relation extraction, LLMs enhance semantic reasoning

System	Task	Functional Axis	Agentic Config	LLM Role	KG Role	Capabilities	Strengths	Limitations
ESCARCOT	Biomedical QA, multi-hop reasoning	LLM-KG Co-operation	Hybrid	Strategy generation, reasoning, GoT code synthesis	Grounding via Cypher queries + vector DB	Graph-of-Thought, tool use, planning	Transparent reasoning, reduced hallucination	Domain-specific, KG dependency
KARMA	KG construction from literature	LLMs for KGs	Hybrid (Multi-agent)	Entity/relation extraction, schema alignment, validation	Target KG construction and refinement	Multi-agent collaboration, conflict resolution	Scalable, consistent KG building	High coordination complexity
ODA	Multi-hop QA, relation extraction	KG-enhanced LLM	Hybrid	Iterative reasoning, memory updates	Subgraph extraction for grounding	Observe-act-reflect loop	Strong multi-hop reasoning	Higher latency due to iterations
KG-Agent	Complex KG reasoning, tool-based QA	LLM-KG Co-operation	Hybrid	Decision-making (tool selection, navigation)	Graph navigation, factual reasoning	Autonomous planning, tool orchestration	Efficient, works with smaller LLMs	Needs structured KG + tools

Table 1. Compact comparison of key LLM-KG Agentic Systems from the Literature Review

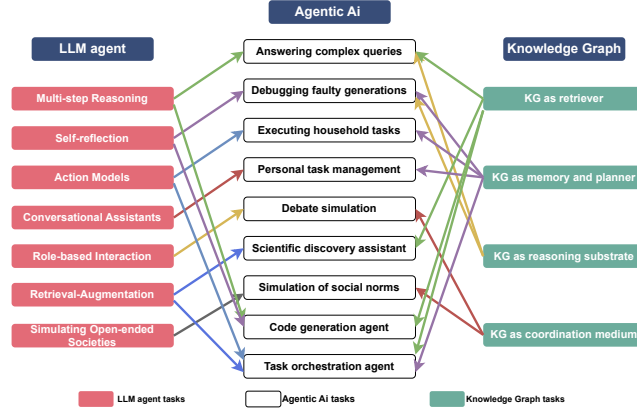


Fig. 1. Interplay between large language model agents and knowledge graphs for complex tasks suited for agentic AI.

Category	Representative Methods	Design Trade-offs	Scalability	Evaluation Settings
Entity Extraction	PromptNER, UniversalNER	Few-shot (low data need) vs. instruction-tuned (higher precision)	High (no fine-tuning)	Domain NER benchmarks, F1-score
Relation Extraction	GPT-RE, REBEL, DeepStruct	Zero/few-shot (fast) vs. CoT-enhanced (better semantics)	Medium (prompt length limits)	RE datasets, precision/recall
KG Completion	KG-S2S, GenKGC, KICGPT	Generative (creative) vs. training-free (low cost)	High for zero-shot	WN18RR / FB15k-237, Hits@1/10
KG Reasoning	LARK, RoG, KG-GPT	Hybrid planning (verifiable) vs. LLM hallucination risk	Low (graph traversal)	Multi-hop QA, accuracy

Table 2. LLMs for KGs Comparison

across supervised, few-shot, and zero-shot settings. Approaches like GPT-RE [66], REBEL [6], DeepStruct [68], and CoT-enhanced few-shot learning [74] reduce data dependence and improve interpretability and generalization. KG reasoning benefits from hybrid LLM–KG frameworks such as LARK [9], RoG [45], and KG-GPT [35], which integrate LLM reasoning with KG-based planning and retrieval to reduce hallucinations and enable logical inference. For KG completion, classical embedding models (TransE [73], DistMult [79], ComplEx [65]) are extended by text-enhanced approaches like KG-BERT [82], StAR [67], and SimKGC [70], as well as generative methods such as KG-S2S [8] and GenKGC [77]. Recent training-free approaches like KICGPT [75] reduce reliance on fine-tuning. Additionally, LLMs contribute to KG embedding, validation, and inconsistency detection (e.g., ChatRule [44]), supporting tasks such as entity alignment, fact-checking, and rule generation, while highlighting trade-offs between flexibility, precision, and scalability.

Category	Representative Methods	Design Trade-offs	Scalability	Evaluation Settings
Knowledge Injection	K-BERT, Sem-K-BERT, KCF-NET	Pretraining integration (accurate) vs. noise from unfiltered KG	Medium (retrain needed)	GLUE/NLI, accuracy
RAG Variants	Naive/Advanced/Modular RAG, GraphRAG, KnowledgeGPT	Static indexing (fast) vs. modular KG retrieval (adaptable)	High (index once)	OpenQA, ROUGE/BLEU

Table 3. KG-Enhanced LLMs Comparison

4 KG-enhanced LLM

Approaches such as K-BERT [40], Sem-K-BERT [78], and KCF-NET [17] enhance language models by integrating knowledge graph information into text representations, improving performance across NLP tasks. Additional methods incorporate external semantics, such as concept-enhanced pretraining [31] and dictionary-based enrichment in Dict-BERT [83]. To address hallucinations and knowledge limitations, Retrieval-Augmented Generation (RAG) [16] introduces external knowledge into LLMs through retrieval, generation, and augmentation, evolving into Naive, Advanced, and Modular variants, with Naive RAG following indexing, retrieval, and generation steps.

Graph RAG [13] generates a summary over a knowledge graph that has been constructed by the LLM from the source data. Table 3 compares knowledge injection versus RAG-based strategies, revealing trade-offs in retraining overhead versus real-time adaptability, with standardized evaluation metrics across NLP tasks.

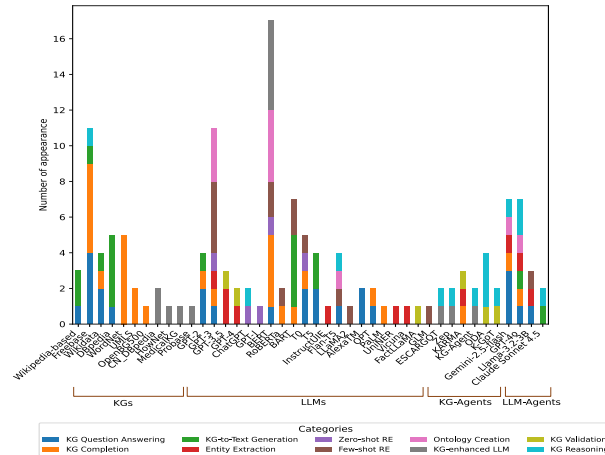


Fig. 2. Statistics of the usage of LLMs and KGs in cited papers per category

5 LLM-KG Cooperation

While the previous sessions considered the paradigms of either KG or LLM as a main component and the other as an external support, this section consider more of a symmetric relationship where LLM and KG are having equal importance. This cooperative dynamics is more significant and neither component is sufficient by themselves alone in this paradigm. Here we discuss some key instantiations of LLM-KG cooperation. In multi-hop question generation, it requires the reasoning over multiple sentences. KEGEL is one such model introduced by [39]. This is trying for multi-hop reasoning with the help of KG, to perform like humans do. It is having three stages, being (1) Understanding context using GPT-2, (2) Information Integration using KG and Answer-aware dynamic Graph Attention Network (GAT), (3) Question Generation using a multi-head attention module over enriched latent representations.

Complex or Multi-hop Question Answering: Complex question answering over KGs involves multiple subjects, interconnected relations, numerical reasoning, and multi-hop inference [37, 81, 14]. Cao and Liu [7] proposed ReLMKG, a multi-hop KBQA method that fuses implicit knowledge from pre-trained language models with explicit KG structure. Similarly, For improving the reasoning Sen et. al. [60] combine a KGQA model with a language model and Baek et. al. [3] retrieve relevant KG facts using similarity metrics and augment LM prompts with these facts to boost answer quality.

Query Generation from text: has shown how effectively is LLMs generating natural language-to-query translations, particularly SPARQL or Cypher for KGs. Converting the text to query is challenging because the model has to perform error-free for both question understanding and for the KG structure [58]. To tackle this, Rony et. al. [58] introduced SGPT, an embedding method that encodes linguistic cues and leverages a pre-trained language model to produce SPARQL queries. And, an enhancement to SPARQLGEN, proposed by [36], supplies the LLM with a single prompt containing the question, an RDF sub-graph. Furthermore, [56] enhanced SPARQLGEN with additional data sources and refined prompting strategies to improve query generation.

Querying LLMs with SPARQL [59] explored the possibility of querying LLMs with SQL. This initiates new directions for research exploration. While SQL is a powerful tool for structured datasets with defined schemas, it falls short in immediately handling information presented as unstructured text. The statement discusses an illustration of a Database-First (DB-first) approach that harnesses the capabilities of LLMs along with traditional Database Management Systems (DBMSs). This combination is utilized to establish a hybrid query execution environment, suggesting an integration of LLMs with conventional database systems for enhanced performance. This concept also suggest the possibility of querying LLMs with SPARQL, which might reveal hidden relations in unstructured data.

KG Chatbots [51] conducted a detailed examination and comparison between conversational AI models like ChatGPT and traditional Question-Answering

Systems (QASs) designed for KGs and suggested merging the attributes of both with the goal of creating KG Chatbots.

6 Agentic AI for LLM+KG

Agentic systems can be categorized based on their reliance on LLMs and KGs: LLM-only systems depend on parametric memory and internal reasoning, while KG-only systems rely on symbolic reasoning and rule-based logic. The most effective approach is hybrid, where agents integrate both paradigms through multipurpose or multi-agent collaboration frameworks.

Language models excel in classification, abstraction and fluency in Natural Languages. However, they suffer from hallucination, lack of persistent memory and the lack of factual grounding. In contrast, Knowledge Graphs can provide structured, high fidelity representations of knowledge, enabling logical inference and verifiability, but they lack the flexibility and contextual understanding of language models. Figure 3 has shown the revolution of KGs, LLM and Agentic AI.

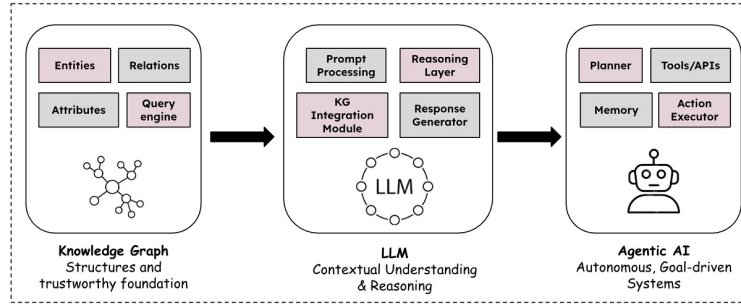


Fig. 3. LLM+KG with Agentic AI.

Agentic systems integrate LLMs and knowledge graphs as complementary components: LLMs handle language understanding, task decomposition, reasoning, and decision-making, while KGs provide structured memory, knowledge organization, and reasoning support. This results in adaptive architectures where planning, reasoning, and knowledge use evolve dynamically over time.

Figure 1 highlights the integration of LLMs and KGs within agentic architectures, where tasks once handled independently are now combined. To explore this design space, we first examine LLM agents, followed by KG-enhanced LLM agents and broader agentic solutions.

As shown in Figure 1, Agentic AI unifies neural (LLMs) and symbolic (KGs) systems: LLMs provide reasoning and action capabilities, while KGs offer grounding through memory and retrieval. Their combination enables advanced applications such as scientific assistance and complex workflow management.

Our taxonomy of LLM agents, inspired by [55], is structured around three dimensions: Reasoning, Acting, and Interacting, covering aspects such as Chain-of-Thought, retrieval, self-reflection, tool use, robotics, and multi-agent coordination. We further propose a taxonomy for KG-enhanced agentic systems based on KG roles, including knowledge retriever, reasoning substrate, memory, and coordination medium.

6.1 LLM agents

LLMs are evolving from passive responders to autonomous agents capable of reasoning, acting, and interacting. Unlike traditional models, agentic systems can make decisions, retrieve information, reflect, and communicate with partial autonomy, drawing from areas such as symbolic reasoning, reinforcement learning, and robotics. Organizing these capabilities into reasoning, acting, and interacting enables the development of increasingly intelligent systems that can also generate new training data, fostering continuous improvement [55].

Reasoning: Agentic LLMs decompose complex problems into intermediate steps through multi-step reasoning instead of single-shot answers [76]. Techniques such as Chain-of-Thought, Tree-of-Thought, Graph-of-Thought, Program-of-Thought, and Self-Consistency enable exploration of solution paths and reduce uncertainty [5]. Program-aided methods further improve accuracy by converting reasoning into executable code, especially for tasks involving math, logic, and structured decision-making.

Self Reflection: Reflexion, Self Refine, and Critic-Actor loops are the reflective methods administered by intelligent agents. This allow agents to critique outputs, detect inconsistencies, and iteratively improve solutions [62]. These mechanisms may trigger automatically based on uncertainty, and the resulting traces can serve as synthetic training data as well. This strengthen the agent’s autonomy and reliability.

Retrieval Augmentation: When external knowledge is needed, RAG enables LLM agents to retrieve information from sources such as documents, APIs, databases, or knowledge graphs, grounding reasoning in up-to-date facts. Techniques like hybrid retrieval, iterative retrieval, query rewriting, and retrieval-chain reasoning improve relevance and accuracy during problem solving [63].

Acting – Action Models: Agentic LLMs interact with environments through action and world models, including vision-language-action systems (e.g., RT-2, OpenVLA, PaLM-E) that link perception to decision-making.

Robots and Tools: LLMs act as planners orchestrating tools, APIs, and agents (e.g., ReAct, AutoGPT, AutoGen), enabling multi-step workflows and coordinated multi-agent collaboration.

Assistants: Domain-specific assistants leverage specialized data and tools to outperform general models in professional tasks.

Interacting – Social Capabilities: LLMs adapt to user emotions and context, with methods like ECoT/CoET supporting aligned responses, including therapeutic applications.

Role-based Interaction: Multi-agent systems assign structured roles (e.g., planner, executor), improving coordination and solution quality (e.g., CAMEL, RoleGPT).

Simulating Open-Ended Societies: Large-scale agent simulations (e.g., Generative Agents, Smallville, Cicero) model social dynamics, cooperation, and complex behaviors across domains.

6.2 KGs in Agentic AI

KGs in agentic architectures serve key roles—retriever, reasoning substrate, memory, and coordination medium—aligning with LLM-KG integration paradigms. As retrievers, they provide structured grounding; as reasoning substrates, they support validation of intermediate steps; and as memory or coordination layers, they enable long-term state management and multi-agent collaboration.

Knowledge retriever:

Knowledge retrievers connect LLM agents to external sources, dynamically fetching relevant structured and unstructured information to support real-time reasoning, fact-checking, and context-aware decision making.

Retrieval Mechanisms and Methodologies: Modern retrieval mechanisms combine dense retrieval (Dense Passage Retrieval) [34], sparse and hybrid methods (BM25 [57] + neural), and learned sparse models like SPLADE [15], with extensions to multimodal retrieval for richer contextual reasoning. While retrievers enable scalability, real-time knowledge access, and domain adaptability, they also introduce challenges such as retrieval errors, latency, bias, and retrieval-generation mismatch, requiring careful query design and filtering.

Reasoning substrate: Graph-of-Thought (GoT) structures reasoning as interconnected graphs, enabling multi-hop validation and parallel exploration over knowledge graphs. This can evolve into Knowledge-Graph-of-Thought (KGoT), where reasoning becomes a dynamic graph combining symbolic and parametric knowledge.

Memory and planning: Knowledge graphs act as external memory for LLM agents, supporting long-term coherence, episodic recall, belief updating, and planning. Systems like Zep (Graphiti engine), AirGraph, and MemoriesDB illustrate how structured, temporal knowledge enables adaptive behavior.

Coordination medium: Shared knowledge graphs serve as coordination substrates in multi-agent systems, allowing agents to interact through a common

semantic space. Similar to blackboard architectures, KG-based coordination enhances collaboration with semantic querying, conflict detection, and structured progress tracking.

6.3 Open Challenges for LLM+KG

Further research is needed to reliably integrate external knowledge into large language models (LLMs) and to develop smaller, more efficient models that retain strong reasoning capabilities. Current approaches explore the use of compact models for lightweight reasoning tasks, offloading only the most complex reasoning steps to larger models to accelerate inference without sacrificing accuracy [1].

Incorporating knowledge from KGs in LLMs: A promising direction for future work lies in seamlessly incorporating knowledge from knowledge graphs (KGs) directly into the inference process of LLMs, thereby eliminating the need to encode factual knowledge in the model’s parameters during training [49]. This would significantly reduce parameter count and model size.

Reducing Training Costs and Model Size by Restricting to Non - Redundant Language Constructs: Training LLMs on minimal, non-redundant subsets of programming or query languages - such as core fragments of SPARQL [21] and XPath [23, 19] - could further decrease the required number of parameters. Another strategy involves generating code or queries in a restricted set of canonical languages and automatically transforming them into the target language, leveraging well-established transformations (e.g., between XML and Semantic Web formats and their associated query languages [10–12, 25, 24, 22]). Additional efficiency gains may come from curating training data to include only satisfiable queries that yield results [18, 20, 26, 19], or by augmenting LLMs with explicit rules through rule-guided prompting [71, 69, 72, 38, 50] or symbolic solvers [53, 50].

Smaller Models through KGs: Reducing LLM parameter counts directly cuts energy use and carbon emissions. A sharp separation between factual knowledge (stored reliably in external knowledge graphs) and pure language understanding (trained on a minimal, high-quality, knowledge-free corpus) is highly desirable. Finding this optimal corpus is a challenging open research problem. Such separation would eliminate or drastically reduce the need for domain-specific fine-tuning and enable new applications like personal KG-enhanced LLMs that mimic an individual’s writing style from their emails/chats while grounding answers in a private personal KG.

6.4 Open Challenges for Agentic AI for LLM+KG

Agentic AI with the goal of Artificial General Intelligence (AGI): A major direction pursues brain-inspired architectures to reach AGI—capable of any intellectual task humans or animals perform [61]. In these systems, LLMs (or their successors) may become auxiliary components in large multi-agent systems for input processing and output verbalisation, while knowledge graphs remain

the primary, efficient, and trustworthy mechanism for long-term knowledge storage.

Novel tasks and applications: In the state-of-the-art, researchers considered how KG tasks can be improved by LLMs and Agentic AI, how LLM tasks can be improved by utilizing the KG and Agentic AI technologies, and how KGs can help in the Agentic AI (for example for offering memory to agents). There is still much future work left for developing novel types of tasks and applications by synergized utilization of LLMs, Agentic AI, and KGs for helping humans in their daily lives.

Utilizing parallelism: Parallel data processing leverages multiple forms of parallelism to improve efficiency: pipeline parallelism processes data in stages across compute units, data parallelism splits workloads into independent fragments, and I/O parallelism distributes data across storage devices to enable faster and concurrent access. For neural networks, model parallelism allows for distributing a model – that is too large for a single device – over several devices, and tensor parallelism for a fine-grained parallel processing of tensor operations. Future research may cover to investigate utilizing these types of parallelisms for processing agents for superior performance. Different hardware is designed for being beneficial for different types of parallelism, i.e., GPUs for data parallelism and FPGAs for pipeline (and data) parallelism [42, 4]. Hence, future work may tackle to investigate how to optimize the processing of agents for different types of hardware like GPUs, FPGAs, and quantum computers, and which hardware is best for processing a certain kind of agent. Furthermore, how to apply data parallelism for the access of different agents to knowledge graphs?

Distributed scenarios: Agentic architectures involve multiple distributed agents communicating via protocols such as Agent2Agent, though challenges remain in agent migration, trust, and efficient task distribution. Current LLM+KG systems often rely on locally stored knowledge graphs, revealing a research gap in distributed KG storage for improved performance and reduced communication costs. Key challenges include distributing agents and tasks effectively, ensuring synchronization, and managing parallel execution. Initial solutions like [85] adapt database ACID properties to coordinate multi-agent systems and ensure reliable processing.

Security and Privacy: Agentic architectures, particularly in distributed settings, raise significant security and privacy concerns. While secure communication protocols can protect interactions, trust in local and remote agents remains challenging, as they may process sensitive data in unintended ways. Although similar issues exist in traditional software, Agentic AI introduces heightened risks due to its autonomy and proximity to user data, requiring careful design to avoid past security pitfalls and ensure privacy protection.

7 Conclusion

This survey provides the first comprehensive exploration of the convergence of Large Language Models (LLMs), Knowledge Graphs (KGs), and the emerging

paradigm of Agentic AI. By systematically categorizing the interplay between these technologies along three functional axes, LLMs for KGs, KG enhanced LLMs, and LLM–KG cooperation, and further extending the analysis to agentic configurations, we establish agentic AI as the crucial architectural and conceptual bridge that unifies symbolic and neural intelligence.

We observe that while LLMs bring generative fluency and adaptive reasoning, and KGs offer verifiable structure and factual grounding, it is the autonomous, goal driven agent layer with its capabilities of planning, tool use, reflection, memory, and multi agent coordination that transforms static hybrid systems into dynamic, trustworthy, and truly intelligent ecosystems.

The introduction of agentic AI fundamentally reshapes the design space of LLM–KG integration. Rather than treating KGs merely as retrieval substrates or LLMs as passive reasoners, agentic frameworks position KGs as active components serving multiple roles simultaneously: knowledge retrievers, reasoning substrates (Graph-of-Thought or Knowledge-Graph-of-Thought), persistent memory stores, and coordination media for multi agent collaboration. Exemplary systems such as ESCARGOT, KARMA, ODA, and KG-Agent, etc demonstrates the practical viability of this synergy. Looking forward, substantial open challenges remain that will shape the future trajectory of this field. Key research directions include developing more efficient integration methods that incorporate KG knowledge directly during inference, reducing model size and training costs through non redundant language constructs and curated training data. Research must also tackle the optimization of agent processing for different hardware. Achieving these goals will require innovations in scalable KG storage, efficient reasoning, parallel and distributed agent execution, and trustworthy multi-agent cooperation ultimately paving the road for the next generation of hybrid neural-symbolic artificial intelligence in Agentic AI.

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